

# AlignAtt without Cross-Attention: Engine-Native Attention Probing for Simultaneous LLM Translation

## Abstract

We study AlignAtt for simultaneous translation with decoder-only LLMs, where the model exposes no decoder cross-attention and modern inference stacks fuse attention inside graph-compiled kernels. We recover source-conditioned provenance from causal self-attention by combining a deterministic prompt layout, offline selection of alignment heads, and exact selective reconstruction of the draft-to-source attention block via *qk-fast*. We then describe an engine-native observer that captures the required query and key tensors under graph-compiled inference without breaking compilation or altering model outputs. The result is a single AlignAtt substrate that applies to both ASR and MT roles while preserving numerical parity with eager reference execution.

## 1 Positioning

Attention-based policies have already proved useful for simultaneous speech translation. AlignAtt derives audio-to-translation alignments from encoder-decoder cross-attention at inference time [1], while StreamAtt addresses long-form direct streaming speech translation through attention-based audio-history selection [3]. Recent interpretability work on multilingual LLMs further identifies sparse translation-specific alignment heads in decoder-only models [2]. Our setting differs from all three: we study a cascade whose MT component is a decoder-only LLM with no cross-attention, and whose deployed runtime executes fused, graph-compiled attention.

We therefore ask whether AlignAtt can be reformulated so that it remains exact, selective, and observable under these constraints. Our answer has two coupled parts: a virtual source-restricted view of self-attention, and an engine-native capture mechanism that exposes the query/key tensors required for selective reconstruction without perturbing model outputs. For evaluation we use Simulstream, a framework

for long-form revision-capable streaming speech translation [4], and we report compute-unaware LongYAAL on unsegmented audio [5]. All experiments use a synchronous cascade loop in order to isolate policy behaviour from scheduler effects. The ASR and MT components are architecturally separable and could in principle be pipelined in future system work, but such overlap is not evaluated here.

## 2 Simultaneous Cascade Architecture

### Streaming input and alignment backend.

The system is invoked on a monotone sequence of timestamps  $t_1 < t_2 < \dots$  with  $t_{k+1} - t_k = \Delta_{\text{chunk}}$  (by default  $\Delta_{\text{chunk}} = 850$  ms). At each  $t_k$ , an alignment backend returns a partial transcript  $\mathbf{s}^{(k)} = (s_1^{(k)}, \dots, s_{N_k}^{(k)})$  with per-word end-times  $\theta_i^{(k)} \in \mathbb{R}_{\geq 0}$ ,  $i = 1, \dots, N_k$ . The analysis below only assumes this contract: an incremental recognition/alignment module that returns a live source prefix together with per-word end-times. Throughout, bold symbols such as  $\mathbf{s}^{(k)}$  denote ordered token sequences, while calligraphic symbols denote index sets; in particular  $\mathcal{S}^{(k)} := \{1, \dots, N_k\}$  is the local index set of the live source prefix at step  $k$ .

**Source accessibility frontier.** Given an inaccessibility margin  $\tau_{\text{in}} \geq 0$ , the *accessible source index set* at time  $t_k$  is

$$\mathcal{S}_{\text{acc}}^{(k)} = \left\{ i \in \mathcal{S}^{(k)} \mid \theta_i^{(k)} \leq t_k - \tau_{\text{in}} \right\}, \quad (1)$$

with  $N_{\text{acc}}^{(k)} = |\mathcal{S}_{\text{acc}}^{(k)}|$  and the convention that on a final chunk ( $t_k = T$ ) we set  $N_{\text{acc}}^{(k)} = N_k$ . The inaccessible complement is  $\mathcal{S}_{\text{inacc}}^{(k)} = \mathcal{S}^{(k)} \setminus \mathcal{S}_{\text{acc}}^{(k)}$ . These positions are conditioned on but *not* allowed to drive accepted target tokens (Sec. 4).

**Translation backend.** The MT backend maps the live source  $\mathbf{s}^{(k)}$  and the accepted target prefix

$\mathbf{y}_{1:m_k}$  to a draft  $\tilde{\mathbf{y}}^{(k)} = (\tilde{y}_1^{(k)}, \dots, \tilde{y}_{L_k}^{(k)})$  of at most  $L_{\text{draft}}$  new target tokens.

**Acceptance and emission.** An AlignAtt policy  $\Pi$  turns the draft  $\tilde{\mathbf{y}}^{(k)}$  into an accepted suffix  $\mathbf{a}^{(k)}$  of length  $\ell_k \in \{0, 1, \dots, L_k\}$ ,

$$\mathbf{a}^{(k)} = \Pi(\tilde{\mathbf{y}}^{(k)}, \mathcal{S}_{\text{acc}}^{(k)}), \quad (2)$$

so  $\mathbf{a}^{(k)} = (\tilde{y}_1^{(k)}, \dots, \tilde{y}_{\ell_k}^{(k)})$  is the retained draft prefix, and the accepted target prefix is extended monotonically,

$$\mathbf{y}_{1:m_{k+1}} = \mathbf{y}_{1:m_k} \parallel \mathbf{a}^{(k)}, \quad (3)$$

with  $m_{k+1} = m_k + \ell_k$ . The invariant  $m_k$  is non-decreasing; already-emitted target tokens are never retracted.

Figure 1 gives the operational view used in this work: a standalone decoder-only Gemma cascade where each chunk triggers an ASR request and an MT request, the confirmed ASR transcript prefix defines the visible MT source span, and the accepted target prefix is reused as causal target prefill.

**Latency functional.** Let  $\tau_j^{\text{emit}}$  denote the wall-clock emission time of target token  $y_j$  and  $\tau_j^{\text{src}}$  the end-time of the source word responsible for it. The primary latency metric we track is the compute-unaware LongYAAL latency [5]:

$$\text{LongYAAL}_{\text{CU}}(\mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{j=1}^{|\mathbf{y}|} (\tau_j^{\text{emit}} - \tau_j^{\text{src}})_+, \quad (4)$$

with  $(u)_+ = \max(u, 0)$ .

### 3 AlignAtt for Decoder-Only LLMs

**From encoder-decoder AlignAtt to decoder-only LLMs.** In its original formulation, AlignAtt exploits decoder cross-attention in an encoder-decoder speech translation model [1]: each decoder row  $\mathbf{A}_{t,\cdot}^{\text{xattn}}$  is normalized over encoder source positions only, so source provenance is built into the architecture. In a decoder-only LLM, there is no cross-attention: a single token stream of length  $n$  mixes instructions, source, conversation history, and the draft being generated, and every row of causal self-attention  $\mathbf{A}_{t,\cdot}^{\text{self}}$  is normalized over all  $n$  prompt positions. Two things are thus

lost relative to the encoder-decoder setting: (i) the attention row no longer distinguishes source from non-source by construction, and (ii) no head is architecturally reserved for source alignment.

**Prompt-induced source span.** We recover a source-target distinction by *prompt layout* rather than architecture. The MT prompt is assembled deterministically from a fixed system preamble  $\mathbf{p}^{\text{sys}}$ , the live source  $\mathbf{s}^{(k)}$ , a task instruction  $\mathbf{p}^{\text{instr}}$  followed by the accepted target prefix  $\mathbf{y}_{1:m_k}$ , and the draft  $\tilde{\mathbf{y}}^{(k)}$ :

$$\mathbf{p}^{(k)} = [\underbrace{\mathbf{p}^{\text{sys}}}_{\mathcal{P}_1} \parallel \underbrace{\mathbf{s}^{(k)}}_{\mathcal{S}^{(k)}} \parallel \underbrace{\mathbf{p}^{\text{instr}}, \mathbf{y}_{1:m_k}}_{\mathcal{P}_2} \parallel \underbrace{\tilde{\mathbf{y}}^{(k)}}_{\mathcal{D}}], \quad (5)$$

where  $\mathcal{S}^{(k)} = \mathcal{S}_{\text{acc}}^{(k)} \sqcup \mathcal{S}_{\text{inacc}}^{(k)}$  is the live source index set at decode step  $k$ ,  $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$  collects all non-source prompt positions, and  $\mathcal{D}$  is the draft. The sets  $\mathcal{S}_{\text{acc}}^{(k)}$  and  $\mathcal{S}_{\text{inacc}}^{(k)}$  are exactly the accessibility partition of Eq. (1). Let  $\phi^{(k)} : \mathcal{S}^{(k)} \hookrightarrow \{1, \dots, n\}$  map local source indices to absolute prompt positions. Eq. (5) makes the source span explicit in prompt space even though the model still executes ordinary causal self-attention.

**Source-restricted attention row.** Given this layout, the object relevant to alignment is simply the slice of self-attention over source columns: for draft position  $t \in \mathcal{D}$  and source index  $s \in \mathcal{S}^{(k)}$ , we read  $\mathbf{A}_{t, \phi^{(k)}(s)}^{(\ell, h)}$ , layer  $\ell$ , head  $h$ . Unlike in the encoder-decoder setting,  $\sum_{s \in \mathcal{S}^{(k)}} \mathbf{A}_{t, \phi^{(k)}(s)}^{(\ell, h)} < 1$  in general, because the row is normalized over the full prompt. The residual mass on non-source prompt positions and on the draft is not noise to be discarded; in Sec. 4 it is kept available during replay, while the source-side decomposition drives the acceptance policy.

**What changes and what stays the same.** Relative to standard AlignAtt, the online decision rule stays the same in spirit: a first-failure scan accepts the longest draft prefix whose source-side alignment signal remains on the accessible side of the source frontier. What changes is the *substrate* on which that signal is observed. In the encoder-decoder case the alignment row is source-normalized by construction, whereas in the decoder-only case we must carve a source-restricted row out of full-prompt self-attention and explicitly account for the residual mass on the accepted target prefix, the rest of the prompt,

### One Gemma checkpoint, two sequential AlignAtt requests

Per chunk: ASR request confirms a transcript prefix, then MT request extends the translation from the visible source span.

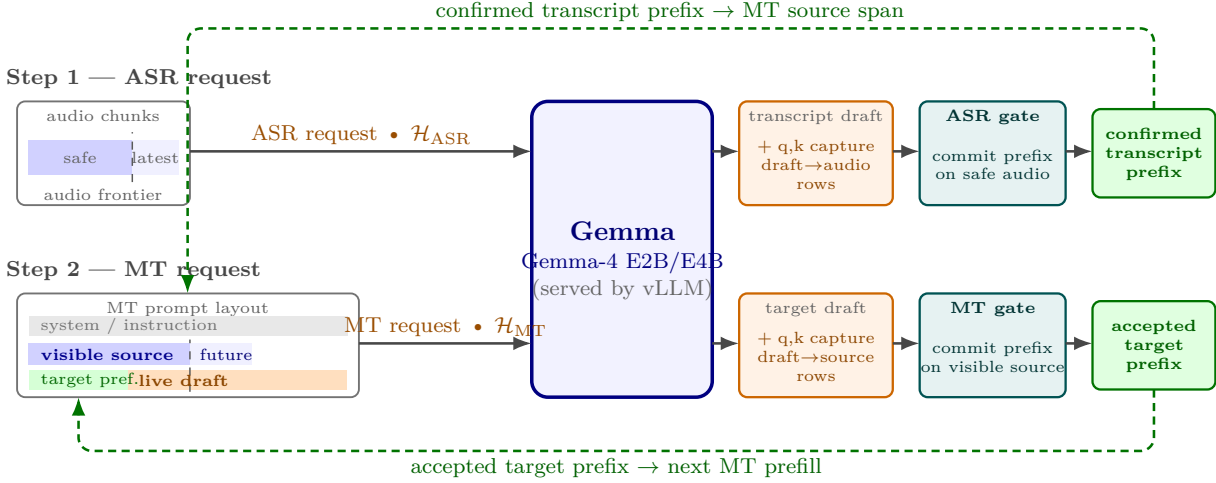


Figure 1: **Operational view of the decoder-only cascade.** A single decoder-only Gemma checkpoint is queried twice per input chunk: first in the ASR role (audio→text) to confirm a transcript prefix, then in the MT role (text→text) to extend the translation from the visible source span. In the runtime this appears as two sequential vLLM requests with role-specific AlignAtt observers ( $\mathcal{H}_{\text{ASR}}$  and  $\mathcal{H}_{\text{MT}}$ ), each reconstructing only the draft-to-source block consumed by its gate. The confirmed ASR transcript prefix feeds the MT source span, and the accepted target prefix is reused as causal target prefill for the next MT step.

and the speculative suffix. Eq. (17) is therefore the decoder-only analogue of the original “do not emit if the token aligns to the last  $f$  source frames” condition, with  $N_{\text{acc}}^{(k)} + b$  replacing the forbidden source tail.

**Alignment heads.** Not all  $LH$  self-attention heads carry alignment signal; many perform syntactic or coreferential roles. We posit the existence of a small head subset  $\mathcal{H} \subset \{1, \dots, L\} \times \{1, \dots, H\}$ ,  $|\mathcal{H}| \ll LH$ , such that the head-averaged source row  $\frac{1}{|\mathcal{H}|} \sum_{(\ell, h) \in \mathcal{H}} \mathbf{A}_{t, \phi^{(k)}(s)}^{(\ell, h)}$  approximates a gold source-to-target alignment on a held-out calibration set  $\mathcal{G} = \{(\mathbf{s}^{(i)}, \mathbf{y}^{(i)}, \mathbf{a}^{(i)})\}_i$ , where  $\mathbf{a}^{(i)} \in \{1, \dots, |\mathbf{s}^{(i)}| \}^{|\mathbf{y}^{(i)}|}$  is a forced word-level alignment. The sparsity assumption is consistent with recent evidence that multilingual LLMs contain a small set of translation-specific token alignment heads [2], although our selection is calibrated directly on the prompt layout and task role used at inference time.

**Offline head selection.** For a single  $(\ell, h)$ , define the argmax alignment

$$\hat{a}_t^{(\ell, h)}(\mathbf{s}, \mathbf{y}) = \arg \max_s \mathbf{A}_{t, \phi(s)}^{(\ell, h)}, \quad (6)$$

and its token-level alignment accuracy on  $\mathcal{G}$ ,

$$\text{Acc}_{\text{align}}(\ell, h) = \frac{1}{|\mathcal{G}|} \sum_i \frac{1}{|\mathbf{y}^{(i)}|} \sum_{t=1}^{|\mathbf{y}^{(i)}|} \mathbf{1}[\hat{a}_t^{(\ell, h)} = \mathbf{a}_t^{(i)}], \quad (7)$$

where  $\hat{a}_t^{(\ell, h)}$  is evaluated at  $(\mathbf{s}^{(i)}, \mathbf{y}^{(i)})$ . We then select  $\mathcal{H} = \text{top-}k_{(\ell, h)} \text{Acc}_{\text{align}}(\ell, h)$  with  $k = 8$  throughout. The calibration corpus  $\mathcal{G}$  is built from forced-alignment outputs on the dev set (MuST-C + in-house).

Figure 2 gives the architecture-aware view used in practice for Gemma-4: the same layer×head score map is overlaid with the full-attention bands, the late shared-KV region, and a reminder that the grid indexes query heads whereas capture cost still follows KV ownership.

**One abstraction, two roles.** The construction of Eqs. (5)–(7) is *role-agnostic*. Let  $r \in \{\text{ASR}, \text{MT}\}$  index the role: for  $r = \text{ASR}$ ,  $\mathbf{s}$  is a sequence of audio-token embeddings and  $\mathbf{y}$  the transcript; for  $r = \text{MT}$ ,  $\mathbf{s}$  is the transcript and  $\mathbf{y}$  the translation. The same LLM weights are used in both roles; only the prompt layout and the head set  $\mathcal{H}$  depend on  $r$ . We write  $\mathcal{H}_r$  for the role-specific head set, and observe empirically that  $|\mathcal{H}_{\text{ASR}} \cap \mathcal{H}_{\text{MT}}| \ll \min(|\mathcal{H}_{\text{ASR}}|, |\mathcal{H}_{\text{MT}}|)$ : the heads that carry audio-to-text alignment are

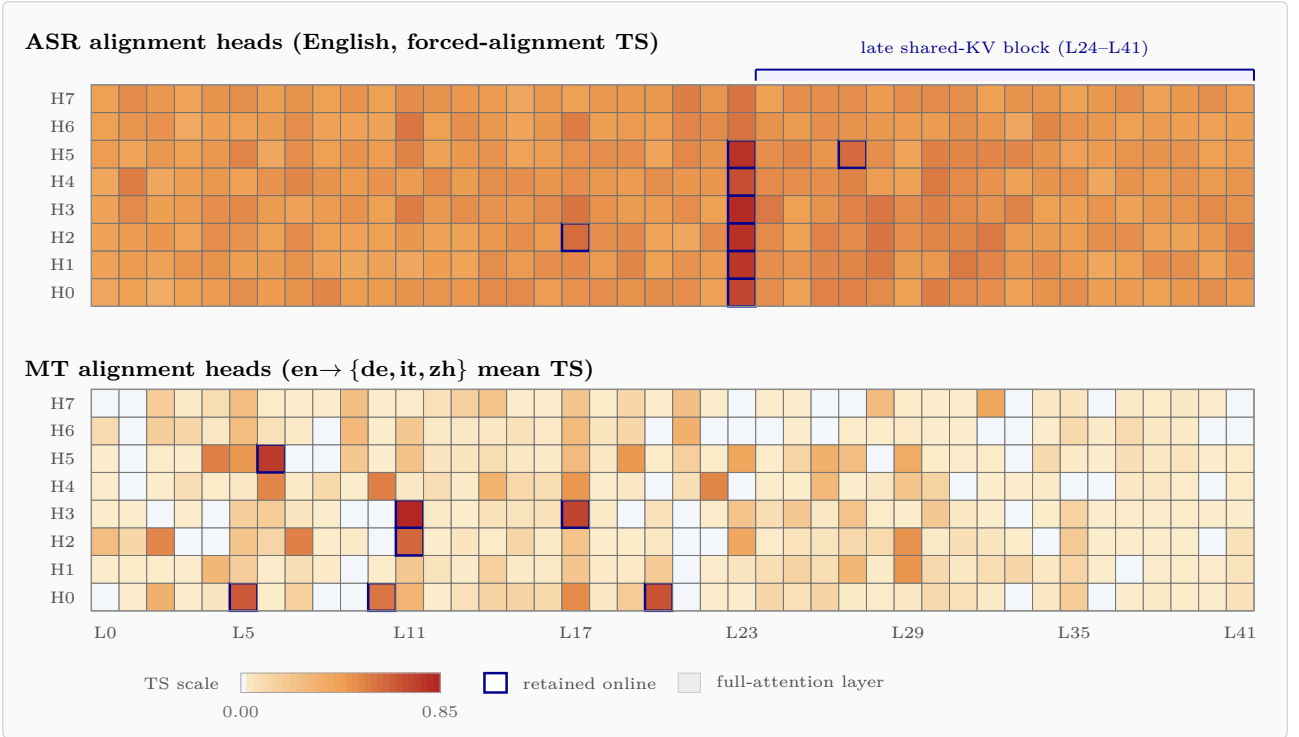


Figure 2: **Architecture-aware view of Gemma AlignAtt heads.** Each heatmap covers Gemma’s 42 layers (horizontal axis) and 8 query heads (vertical axis); color encodes mean token-alignment score TS on a shared  $[0, \max]$  scale, so empty cells read as low rather than “below threshold”. *Top*: audio-to-text alignment heads, scored by forced-alignment MAE on English dev audio. *Bottom*: source-to-target translation heads, averaged over  $\text{en} \rightarrow \{\text{de}, \text{it}, \text{zh}\}$ . Blue outlines mark the per-role retained heads ( $\mathcal{H}_{\text{ASR}}$  and  $\mathcal{H}_{\text{MT}}$ ); grey vertical stripes mark full-attention layers; the blue bracket delimits the late shared-KV block, where observer cost follows KV-group ownership rather than query-head count. The two retained sets concentrate in disjoint depth regions, consistent with per-role functional specialization.

largely distinct from those that carry source-to-target alignment, consistent with per-role functional specialization. The abstraction is the same in both roles; what remains is to realize it operationally. Sec. 4 shows how to reconstruct only the draft-to-source slice needed by the policy, and how to obtain the required query/key tensors under graph-compiled inference where the native attention matrix is never materialized.

## 4 Selective Reconstruction and Engine-Native Capture

**Method overview.** Section 3 defined the object we want: source-restricted self-attention on a small set of alignment heads. The method has two coupled parts. First, we reconstruct exactly only the draft-to-source attention block that the acceptance policy consumes. Second, we make the required query/key tensors observable inside a fused, graph-compiled runtime where the full

attention matrix never appears in memory.

### 4.1 Selective Reconstruction for Acceptance

**Per-head attention and the policy-visible block.** At decoding step  $k$ , let  $\mathbf{p}^{(k)}$  be the prompt of Eq. (5), with local source index set  $\mathcal{S}^{(k)} = \mathcal{S}_{\text{acc}}^{(k)} \sqcup \mathcal{S}_{\text{inacc}}^{(k)}$ , non-source context  $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ , and draft  $\mathcal{D}$ , and prompt-position map  $\phi^{(k)} : \mathcal{S}^{(k)} \hookrightarrow \{1, \dots, n\}$ . Here  $\mathcal{S}_{\text{acc}}^{(k)}$  and  $\mathcal{S}_{\text{inacc}}^{(k)}$  are the accessible and inaccessible parts of the live source at the current decode step. For each attention layer  $\ell \in \{1, \dots, L\}$  and head  $h \in \{1, \dots, H\}$ , let  $\mathbf{Q}^{(\ell, h)}, \mathbf{K}^{(\ell, h)} \in \mathbb{R}^{n \times d_\ell}$  be the post-RoPE, post-norm query and key tensors. The single-head attention row at draft position  $t \in \mathcal{D}$  is

$$\mathbf{A}_{t,\cdot}^{(\ell, h)} = \text{softmax}\left(\mathbf{Q}_t^{(\ell, h)} (\mathbf{K}^{(\ell, h)})^\top + \mathbf{m}_t\right), \quad (8)$$

with  $\mathbf{m}_t$  the causal (optionally sliding-window) mask. Gemma-4 folds the usual  $1/\sqrt{d}$  scaling

into  $\mathbf{Q}$ . Fused kernels compute  $\mathbf{A}^{(\ell,h)}\mathbf{V}^{(\ell,h)}$  without materializing  $\mathbf{A}^{(\ell,h)}$ .

**qk-fast selective reconstruction.** Let  $\mathcal{H} \subset \bigcup_{\ell}\{\ell\} \times \{1, \dots, H\}$  be the AlignAtt head set, with  $|\mathcal{H}| \ll LH$ , selected offline from the alignment-head calibration corpus. The acceptance policy never consumes the full  $n \times n$  matrix: it only needs rows indexed by the draft and columns indexed by the source span. We therefore reconstruct only that block:

$$\hat{\mathbf{A}}_{t,s}^{(\ell,h)} = \frac{\exp(\mathbf{Q}_t^{(\ell,h)} \cdot \mathbf{K}_{\phi^{(k)}(s)}^{(\ell,h)})}{\sum_{j=1}^n \exp(\mathbf{Q}_t^{(\ell,h)} \cdot \mathbf{K}_j^{(\ell,h)} + m_{t,j})}, \quad (9)$$

for  $t \in \mathcal{D}$ ,  $s \in \mathcal{S}^{(k)}$ , and  $(\ell, h) \in \mathcal{H}$ . Eq. (9) recovers *exactly* the same source-reindexed softmax block as Eq. (8); the only discrepancy is floating-point reassociation between the fused kernel and our out-of-band recomputation. The denominator still ranges over all  $n$  prompt positions, so the runtime must expose  $\mathbf{K}^{(\ell,h)}$  for the full prompt even though the reconstructed object is indexed only by source positions  $s \in \mathcal{S}^{(k)}$ . Figure 3 illustrates the construction.

**From reconstructed rows to provenance features.** The averaged source row for token  $t$  is

$$\bar{\mathbf{r}}_t(s) = \frac{1}{|\mathcal{H}|} \sum_{(\ell,h) \in \mathcal{H}} \hat{\mathbf{A}}_{t,s}^{(\ell,h)}, \quad s \in \mathcal{S}^{(k)}, \quad (10)$$

and the corresponding source-side provenance scores are

$$\pi_t^{\text{acc}} = \sum_{s \in \mathcal{S}_{\text{acc}}^{(k)}} \bar{\mathbf{r}}_t(s), \quad (11)$$

$$\pi_t^{\text{inacc}} = \sum_{s \in \mathcal{S}_{\text{inacc}}^{(k)}} \bar{\mathbf{r}}_t(s). \quad (12)$$

These two quantities partition the total source mass according to the accessibility frontier. Residual mass on non-source prompt positions and on the draft is not used by the acceptance rule and is recovered only in the full prompt-space replay of Sec. 4 when diagnostic accounting is needed. The raw averaged row  $\bar{\mathbf{r}}_t$  is useful as a source-mass feature, but the source argmax itself is more stable when computed from a per-head normalized representation. Let  $\mathbf{r}_t \in \mathbb{R}^{|\mathcal{H}| \times N_k}$  stack the per-head source rows,

$$[\mathbf{r}_t]_{(\ell,h),s} = \hat{\mathbf{A}}_{t,s}^{(\ell,h)}.$$

**Prefix-online Welford normalization.** To make the per-token argmax robust to head-scale drift along the draft, we normalize each row against the running moments  $(\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t^2)$  of all prior rows, maintained by a standard prefix-online Welford recurrence over  $\mathbf{r}_t \in \mathbb{R}^{|\mathcal{H}| \times N_k}$ :

$$\tilde{\mathbf{r}}_t = (\mathbf{r}_t - \boldsymbol{\mu}_t) \oslash \sqrt{\boldsymbol{\sigma}_t^2 + \varepsilon}, \quad \varepsilon = 10^{-6}. \quad (13)$$

We then apply a 1-D median filter of width  $w$  independently to each head along the source axis,

$$u_t^{(\ell,h)}(s) = \text{median}_{|j| \leq (w-1)/2} \tilde{\mathbf{r}}_t^{(\ell,h)}(s+j), \quad (14)$$

with default  $w = 7$ . The aligned source position is then

$$\hat{s}_t = \arg \max_s \bar{u}_t(s), \quad \bar{u}_t(s) = \frac{1}{|\mathcal{H}|} \sum_{(\ell,h)} u_t^{(\ell,h)}(s), \quad (15)$$

over  $s \in \{1, \dots, N_k\}$ .

**Acceptance policy.** The policy  $\Pi$  of Eq. (2) is a first-failure scan over the draft indices  $t = 1, 2, \dots, L_k$  with four stop conditions. Let  $\hat{s}_{t-1}$  denote the previous accepted alignment (undefined at  $t = 1$ ), let  $\rho \in \mathbb{N}$  be the rewind threshold (default 3), and  $b \in \mathbb{N}$  the border margin (default 0):

$$\text{REWIND} : \hat{s}_{t-1} - \hat{s}_t > \rho, \quad (16)$$

$$\text{SOURCE-FRONTIER} : \hat{s}_t \geq N_{\text{acc}}^{(k)} + b, \quad (17)$$

$$\text{ARGMAX-MASS-WEAK} : \bar{\mathbf{r}}_t(\hat{s}_t) < \tau_{\text{argmax}}, \quad (18)$$

$$\text{PROVENANCE-WEAK} : \pi_t^{\text{acc}} < \tau_{\text{src}}, \quad (19)$$

all evaluated in order. The accepted length is the largest  $\ell_k$  such that none of Eq. (16)–(19) fires for any  $t \leq \ell_k$ . Finally,  $\ell_k$  is rounded back to the nearest stability-unit boundary (end of a whitespace- or punctuation-delimited word), so partial subword fragments are never emitted.

**Monotone source-side gating under frontier refinement.** We isolate the precise part of the policy that is monotone in the accessibility frontier.

**Proposition 1** (Frontier-monotone source-side gates). *Fix a draft token  $t$  and its reconstructed source row  $\bar{\mathbf{r}}_t$  on a fixed live source index set  $\mathcal{S}^{(k)}$ . Suppose the accessibility partition is refined by moving source indices from  $\mathcal{S}_{\text{inacc}}^{(k)}$  to a larger accessible set  $\mathcal{S}'_{\text{acc}} \supseteq \mathcal{S}_{\text{acc}}^{(k)}$ , with  $N'_{\text{acc}} := |\mathcal{S}'_{\text{acc}}|$ , while leaving  $\bar{\mathbf{r}}_t$  unchanged. If  $\hat{s}_t < N_{\text{acc}}^{(k)} + b$  (Eq. (17))*

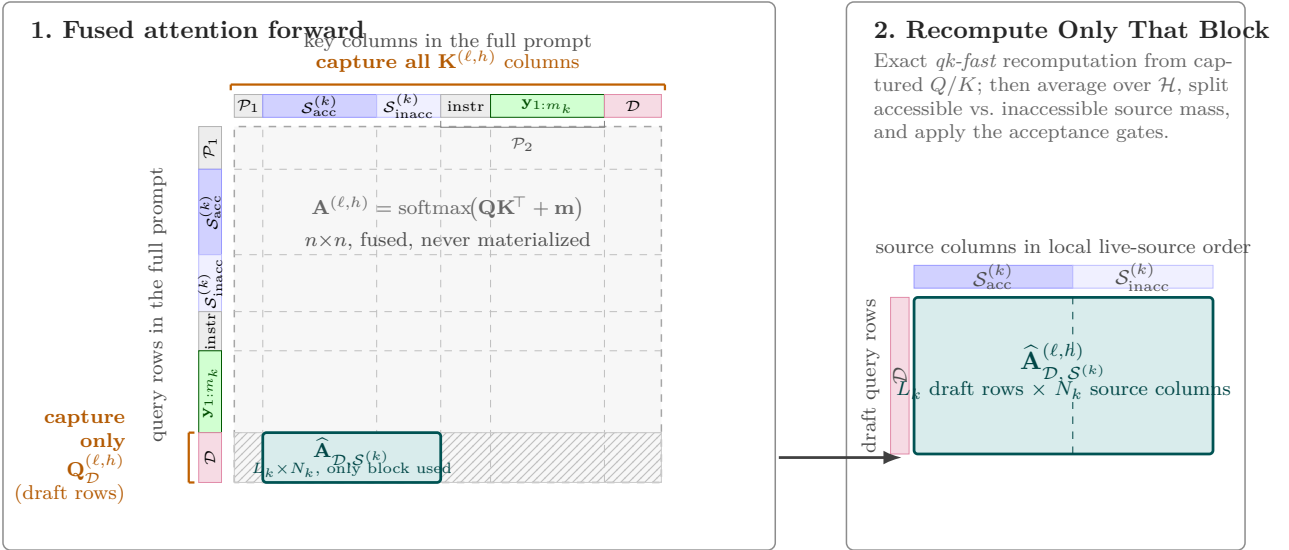


Figure 3: **Selective reconstruction with engine-native capture.** The  $n \times n$  attention matrix  $\mathbf{A}^{(\ell,h)}$  is executed entirely inside the fused attention kernel, so the full  $n \times n$  matrix is never materialized. For each AlignAtt head  $(\ell, h) \in \mathcal{H}$ , the observer copies  $\mathbf{K}^{(\ell,h)}$  for all positions and  $\mathbf{Q}^{(\ell,h)}$  only for the draft rows  $\mathcal{D}$  into fixed-shape buffers. The green band  $\mathbf{y}_{1:m_k}$  inside  $\mathcal{P}_2$  marks the committed target prefix that grows across chunks; hatched cells are never reconstructed. After the forward, Eq. (9) recomputes exactly only the draft-to-source block  $\hat{\mathbf{A}}_{\mathcal{D}, \mathcal{S}^{(k)}}^{(\ell,h)}$ , which is the sole object consumed by the source-side provenance scores of Eqs. (11)–(12) and by the acceptance policy of this section.

and  $\pi_t^{\text{acc}} \geq \tau_{\text{src}}$  (Eq. (19)) before the refinement, then both inequalities remain satisfied after the refinement.

*Proof sketch.* The frontier gate Eq. (17) compares  $\hat{s}_t$  against a threshold that increases from  $N_{\text{acc}}^{(k)} + b$  to  $N'_{\text{acc}} + b$ , so once the inequality is satisfied it remains satisfied after the refinement. The provenance gate Eq. (19) depends on the same reconstructed attention row and on the accessible set through Eq. (11); as  $\mathcal{S}_{\text{acc}}^{(k)}$  grows to  $\mathcal{S}'_{\text{acc}}^{(k)}$ ,  $\pi_t^{\text{acc}}$  is non-decreasing. Both source-side gates are therefore monotone under frontier refinement.  $\square$

Proposition 1 is what makes the monotone emission rule of Eq. (3) coherent at the policy level: relabeling source positions from inaccessible to accessible cannot, by itself, flip an already accepted token back to rejected. Any later change in decision must come from a different draft token or a different reconstructed attention row, not from the frontier split alone.

**Robustness to reconstruction error.** Write  $\mathbf{A}_{t,s}^{(\ell,h)} := \mathbf{A}_{t,\phi^{(k)}(s)}^{(\ell,h)}$  for the exact source-reindexed block. Let  $\varepsilon_\infty = \max_{(\ell,h,t,s)} |\hat{\mathbf{A}}_{t,s}^{(\ell,h)} - \mathbf{A}_{t,s}^{(\ell,h)}|$  be the worst-case reconstruction error due to fp-reassociation. Under the median filter of width  $w$  and Welford normalization, a sufficient condition

for the argmax of Eq. (15) to be invariant under replacing  $\hat{\mathbf{A}}$  by  $\mathbf{A}$  is

$$\varepsilon_\infty < \frac{1}{2} (\bar{u}_t(\hat{s}_t) - \max_{s \neq \hat{s}_t} \bar{u}_t(s)), \quad (20)$$

i.e. the argmax margin dominates the reconstruction perturbation. Empirically  $\varepsilon_\infty \leq 1.2 \times 10^{-2}$  on Gemma-4 E4B prompts, which is well below typical argmax margins under Welford-normalised median filtering; accordingly we observe identical policy decisions at every sampled step.

**Complexity.** Materializing  $\mathbf{A}$  over all layers and heads costs  $\Theta(LHn^2)$  per forward. Selective reconstruction costs, per draft step,

$$\mathcal{C}_{\text{qk-fast}} = \Theta(|\mathcal{H}| \cdot |\mathcal{D}| \cdot n \cdot \bar{d}) + \Theta(|\mathcal{H}| \cdot |\mathcal{D}| \cdot N_k \cdot w), \quad (21)$$

with  $\bar{d} \in \{256, 512\}$  on Gemma-4 and the second term accounting for the median filter. For  $|\mathcal{H}|=8$ ,  $|\mathcal{D}| \leq 16$ ,  $n \leq 1024$ ,  $N_k \leq 128$ ,  $w=7$ , the total is  $< 6 \times 10^5$  scalar multiplications per decoding step.

## 4.2 Engine-Native Realization under Compiled Inference

**Why runtime observability is part of the method.** Eq. (9) is only useful if the runtime exposes the selected query and key ten-

sors. Under fused attention kernels (FlashAttention family) and ahead-of-time graph compilation (`cudaGraph=full`), it does not: the Q/K tensors never leave the fused kernel, Python-level recorders cannot hook into the compiled graph, and unused operator outputs are removed by dead-code elimination during the `torch.inductor` lowering pass. The observer below is therefore not ancillary tooling; it is the mechanism that makes selective reconstruction available in the actual inference stack.

**Capture primitive.** We install at import time a per-layer stateless observer with three fixed-shape buffers: one for prompt keys, one for decode queries, and one for decode keys. Let  $\mathcal{H}_\ell = \{h : (\ell, h) \in \mathcal{H}\}$ . Capture is a single static scatter-add per buffer,

$$\mathbf{B}_\star^{(\ell)} += \text{scatter}(\text{pos}_\star(\ell), \mathbf{X}_\star^{(\ell, \mathcal{H}_\ell)}), \quad (22)$$

for  $\mathbf{X}_\star \in \{\mathbf{Q}^{(\ell)}, \mathbf{K}^{(\ell)}\}$ , implemented as `torch.Tensor.scatter_add_` over statically-shaped index tensors. Fixed-shape scatter-adds are compile-safe: they avoid `nonzero()` and Python conditionals. The captured graph therefore stays independent of runtime shapes within one `cudaGraph=full` bucket.

**Compile-visible custom op.** Let  $\Phi : (\ell, \text{pos}, \mathbf{Q}, \mathbf{K}) \mapsto \mathbf{0}$  be the capture operator, registered as `alignatt::capture_mt_qk` with declared purity `mutates_args = ()`. This makes `torch.compile` treat  $\Phi$  as a value-returning operator whose writes to the observer buffers remain outside the alias analyser. This is what allows  $\Phi$  to be fused into the compiled graph rather than rejected as a side-effecting barrier.

**Sentinel dependency against dead-code elimination.** Under `cudaGraph=full`, any call whose return value does not flow into the graph output is eliminated. To prevent elimination without altering numerics, the patched attention forward injects a zero sentinel:

$$\mathbf{o}'_\ell = \mathbf{o}_\ell + \Phi(\ell, \text{pos}, \mathbf{Q}^{(\ell)}, \mathbf{K}^{(\ell)}), \quad (23)$$

with  $\mathbf{o}_\ell = \mathbf{A}^{(\ell)} \mathbf{V}^{(\ell)}$  and  $\Phi \equiv \mathbf{0}$ , which is numerically identical to  $\mathbf{o}_\ell$  but places  $\Phi$  in the transitive fan-in of the graph output, so  $\Phi$  cannot be DCE-pruned. This is the smallest algebraic perturbation sufficient to keep the observer alive under graph compilation.

**Replay consistency.** AOT-compiled graphs resolve attributes by direct lookup. To avoid a first-pass attribute miss after process restart, every per-layer observer slot  $\mathbf{O}_\ell$  is pre-seeded at patch-install time. If layer  $\ell$  has no live observer,  $\Phi$  returns immediately. The capture path is thus gated purely by whether the backend has installed a live observer, without any graph recompilation.

**Offline reconstruction.** At the end of each decode step we read the buffers and replay the draft rows in prompt space, with prompt keys and draft keys stored separately. Writing  $\mathbf{B}_\star^{(\ell, h)} := \mathbf{B}_\star^{(\ell)}[h]$  for brevity,

$$\mathbf{P}^{(\ell, h)} = \mathbf{B}_{\text{dQ}}^{(\ell, h)} (\mathbf{B}_{\text{pK}}^{(\ell, h)})^\top, \quad (24)$$

$$\mathbf{S}^{(\ell, h)} = \mathbf{B}_{\text{dQ}}^{(\ell, h)} (\mathbf{B}_{\text{dK}}^{(\ell, h)})^\top + \mathbf{m}_{\text{causal}}, \quad (25)$$

$$\tilde{\mathbf{A}}^{(\ell, h)} = \text{softmax}[\mathbf{P}^{(\ell, h)}, \mathbf{S}^{(\ell, h)}], \quad (26)$$

with  $\mathbf{m}_{\text{causal}}$  the strict upper-triangular  $-\infty$  mask. Restricting  $\tilde{\mathbf{A}}^{(\ell, h)}$  to the source columns  $\phi^{(k)}(s)$  and reindexing by  $s \in \mathcal{S}^{(k)}$  recovers the block of Eq. (9), and hence  $\tilde{\mathbf{r}}_t$  and the source-side provenance scores of Eqs. (10)–(12). The remaining columns yield the residual mass on  $\mathcal{P}$  and  $\mathcal{D}$  when diagnostic accounting is needed.

**Qualitative reconstruction example.** Figure 4 instantiates the full reconstruction stack on a text-only MT probe. We display the exact prompt partition seen by the decoder, aggregate reconstructed token-level rows to the word level for readability, and keep the four-way provenance accounting visible for every drafted word. The selected snapshot is taken immediately before the AlignAtt gate rejects the next word, so the figure shows both alignment quality and the operational decision boundary.

The green band in Fig. 4 is the already accepted target prefix  $\mathbf{y}_{1:m_k}$  carried over from earlier streaming steps; it is causal context for the decoder, not the object of the frontier gate. The orange words are the current draft  $\tilde{\mathbf{y}}^{(k)}$ , scanned left-to-right by the policy at the current step  $k$ . Thus  $\hat{s}_{t-1}$  in Eq. (16) refers to the previous drafted word *within the same live draft*, not to a token accepted at an earlier streaming step. A drafted word may still attract large self-attention mass on the accepted prefix and yet remain admissible as long as its source-side signal stays inside the accessible frontier; this is exactly why

the residual provenance terms are kept explicit in the decoder-only formulation.

**Residual provenance in decoder-only AlignAtt.** Unlike encoder-decoder AlignAtt, the source-restricted rows of Eq. (??) are not normalized over the source alone: attention can also land on the accepted target prefix, the rest of the prompt template, and the speculative suffix. On the French qualitative probe bank used to mine Fig. 4 (78 scored prefix-continuation probes on an NVIDIA A40), drafted target units allocate on average 9% to accessible source tokens, 8% to still-inaccessible source tokens, 81% to non-source prompt positions (system/template plus accepted prefix), and 2% to the speculative suffix. Restricting to the target units that AlignAtt actually accepts shifts these shares to 14%/1%/83%/2%. These residual masses are therefore not implementation noise but a structural difference from standard AlignAtt.

**Numerical parity.** Let  $\mathbf{A}^{(\text{TF})}$  denote the reference attention tensor produced by an uncompiled Transformers forward on the identical prompt. We observe

$$\|\Delta\mathbf{A}\|_\infty \leq 1.2 \times 10^{-2}, \quad \|\Delta\mathbf{A}\|_1 / (nL_k) \leq 4 \times 10^{-4}, \quad (27)$$

with  $\Delta\mathbf{A} := \tilde{\mathbf{A}} - \mathbf{A}^{(\text{TF})}$ , on a  $167 \times 12$  reference row set, with bit-identical acceptance decisions  $\Pi(\tilde{\mathbf{y}})_{\text{vLLM}} = \Pi(\tilde{\mathbf{y}})_{\text{TF}, \text{eager}}$  across a curated parity set of  $\geq 128$  prompts. Under Eq. (20) the observed  $\|\cdot\|_\infty$  margin is well below typical post-Welford argmax gaps, so the parity is not accidental but a direct consequence of the robustness bound.

**Paper-only seam benchmark.** To isolate capture cost from full SimulStream variability, we benchmark the same fixed text-only prompt suite under three MT seams: a minimal Transformers eager reference, a Transformers *qk-fast* reference, and the engine-native vLLM *qk-fast* path used by the runtime. Each seam runs in its own subprocess, with one unmeasured warmup pass followed by repeated hot runs on the same 16 prompts. Figure 5 reports the median latency per generated token, while the generated JSON artifact keeps the per-length-bin and per-prompt results for appendix tables.

**End-to-end throughput.** With  $\Phi$  fused into the compiled graph, the cost of the observer is

|                           | Qwen3 forced | Gemma AlignAtt |
|---------------------------|--------------|----------------|
| WER ↓                     | 0.113        | 0.269          |
| CER ↓                     | 0.046        | 0.122          |
| Boundary F1 $\pm 3w$ ↑    | 0.878        | 0.042          |
| Mean  boundary lag  (s) ↓ | 0.145        | 0.403          |
| Load time (s)             | 63.7         | 123.0          |
| RTF (wall-clock) ↓        | 0.306        | 0.628          |

Table 1: **ASR alignment comparison on ccpXHNfaoy (360 s).** Qwen3 forced uses a dedicated 3B-parameter aligner; Gemma AlignAtt reuses the decoder-only LLM already needed for MT. Boundary F1 and lag are computed against 47 gold word boundaries with a  $\pm 3$ -word tolerance.

dominated by the scatter-adds of Eq. (22) and the offline  $\mathbf{QK}^\top$  of Eqs. (24)–(25). On the control clip we measure an RTF penalty below 3% relative to the baseline *graph-compiled* runtime.

## 5 Experiments

**Setup.** All experiments run on a single NVIDIA A40 with a graph-compiled vLLM runtime (`cuda-graph=full`) and Gemma-4 E4B-it in bfloat16. The evaluation corpus is the 21-clip IWSLT 2026 in-house dev-set ( $\sim 2.1$  h total, academic research talks), resegmented by OmniSTEval v0.1.7. We report BLEU and chrF against the dev reference, XCOMET-XL [7] for translation adequacy, and the compute-unaware and computation-aware variants of LongYAAL [5] for latency. For detailed ASR diagnostics we use a single 360 s clip (`ccpXHNfaoy`) for which word-level boundary references are available. All cascades use synchronous ASR/MT under the same chunked schedule.

### 5.1 ASR Alignment: Qwen3 Forced vs Gemma AlignAtt

The single-substrate claim of Sec. 3 rests on AlignAtt being usable as the alignment backend in its own right, not only as the MT policy. We therefore compare the Gemma AlignAtt backend against a dedicated forced aligner (Qwen3-ASR [8] followed by the Qwen3 forced aligner) as the ASR component of the cascade, evaluated on the `ccpXHNfaoy` clip with 47 reference word boundaries.

Table 1 shows the expected direction of the specialization gap: the forced aligner produces lower WER (0.113 vs 0.269), tighter boundaries



|                               | Qwen3 forced (LA) | Gemma AlignAtt     | $f$ | $\rho$ | BLEU $\uparrow$ | CU (ms) $\downarrow$ | CA (ms) |
|-------------------------------|-------------------|--------------------|-----|--------|-----------------|----------------------|---------|
| Mean WER $\downarrow$         | 0.092             | —A                 | 4   | 200    | 34.32           | 668                  | 363     |
| Mean CER $\downarrow$         | 0.042             | —B                 | 4   | 50     | 35.44           | 821                  | 368     |
| BLEU $\uparrow$               | 74.82             | —C                 | 5   | 200    | 35.03           | 911                  | 612     |
| chrF $\uparrow$               | 90.81             |                    |     |        |                 |                      |         |
| LongYAAL CU (ms) $\downarrow$ | 865.7             | Q (quality anchor) | 4   | 200    | 44.35           | 3561                 | 3359    |
| LongYAAL CA (ms) $\downarrow$ | 672.7             |                    |     |        |                 |                      |         |

Table 2: **21-clip ASR alignment comparison**, en-en (ASR-as-MT), chunk 800 ms. BLEU/chrF and LongYAAL are corpus-level on the OmniSTEval-resegmented dev-set; WER/CER are per-clip means over the 21 clips. Gemma AlignAtt entries pending re-run.

(0.145s vs 0.403s mean absolute lag), and a higher boundary F1. AlignAtt’s low boundary F1 reflects a segmentation-granularity mismatch — the streaming policy commits at each decoder token rather than at prosodic boundaries — and is therefore not directly comparable without mapping both outputs to the same unit. The key point for the present paper is that a single LLM substrate produces alignment signals strong enough to drive streaming commit decisions at all, at roughly twice the RTF of a dedicated aligner, while removing the need to host a second 3B-parameter ASR model.

To complement the single-clip diagnostic, Table 2 reports aggregate en-en (ASR-as-MT) metrics over the full 21-clip dev-set at chunk size 800 ms. BLEU, chrF, and LongYAAL are corpus-level scores from OmniSTEval; WER and CER are averaged across clips.

## 5.2 Streaming Latency/Quality Frontier

We next treat the same Gemma AlignAtt backend as an end-to-end transcriber and sweep its two policy thresholds — the frame threshold  $f$  (Eq. (17)) and the rewind threshold  $\rho$  (Eq. (16)) — to expose the streaming latency/quality frontier on the full 21-clip dev-set under the en-en (ASR-as-MT) task. Table 3 reports three operating points below 1s of CU-LongYAAL (the regime targeted by this work), plus a quality-anchor point at  $\rho=200$  that trades  $\sim 3.5$ s of CU for +9 BLEU.

Within the sub-1s regime the three configurations trace a narrow band (34.3–35.4 BLEU; 668–911 ms CU), with  $\rho=50$  (config B) offering the best BLEU and  $f=4, \rho=200$  (config A) the lowest CU. The  $\sim 9$  BLEU gap between A–C and the

Table 3: **Gemma AlignAtt en-en streaming frontier** on the 21-clip dev-set ( $k=6$ ). Configurations A–C sit below one second of CU-LongYAAL. Q uses the original (pre-fix) rewind logic at the same  $(f, \rho)$  as A and shows the upper quality bound reachable at multi-second latency.

| Pair  | Regime  | BLEU  | chrF | COMET | LongYAAL (s) |      |
|-------|---------|-------|------|-------|--------------|------|
|       |         |       |      |       | CU           | CA   |
| en→de | low     | 28.76 | 62.1 | 0.875 | 2.00         | 1.63 |
|       | quality | 32.63 | 64.2 | 0.902 | 3.53         | 3.14 |
| en→zh | low     | 36.01 | 35.0 | 0.743 | 1.95         | 1.77 |
|       | quality | 39.86 | 37.8 | 0.778 | 3.27         | 3.09 |
| en→it | low     | 40.10 | 68.0 | 0.805 | 1.98         | 1.62 |
|       | quality | 44.46 | 70.1 | 0.841 | 3.48         | 3.10 |

Table 4: **End-to-end cascade on IWSLT 2026 dev-set** (21 clips, OmniSTEval longform resegmentation; COMET = XCOMET-XL). Low regime uses  $\Delta_{\text{chunk}}=850$  ms and the AlignAtt policy of Sec. 4; quality regime uses  $\Delta_{\text{chunk}}=1500$  ms.

quality anchor Q quantifies the cost of the sub-1s target: without the anchored-reference rewind fix described in Sec. 3, the policy only reaches its upper quality bound at  $\sim 3.6$ s of CU-LongYAAL.

## 5.3 End-to-End Cascade Quality and Latency

Table 4 reports the end-to-end cascade (Qwen3 forced ASR + Gemma AlignAtt MT, chunked synchronously) for the three IWSLT 2026 dev-set target languages. We report a low-latency regime at  $\Delta_{\text{chunk}}=850$  ms and a single quality-anchor row per pair at  $\Delta_{\text{chunk}}=1500$  ms.

At  $\sim 2$ s CU-LongYAAL all three pairs sit in a narrow band, and the quality regime adds  $\sim 1.5$ s of CU for +3–+4 BLEU and +0.02–+0.04 XCOMET-XL. The COMET gap between en-de/en-it ( $\geq 0.81$ ) and en-zh ( $\leq 0.78$ ) is consistent with the well-known distributional difference of Chinese references: word-segmentation choices and the lower chrF ( $\leq 38$ ) magnify the gap even at similar BLEU. Improving the en-zh prompt or introducing target-language domain priors is left

for future work.

## 6 Discussion

### Single-substrate vs specialist cascades.

The comparison in Sec. 5.1 quantifies the engineering cost of removing a dedicated aligner: roughly  $2\times$  WER and  $\sim 0.25$  s of additional boundary lag on the diagnostic clip, against a halving of model footprint and the removal of an entire runtime. In submissions where the MT model is already the dominant cost, reusing its weights for ASR alignment via AlignAtt becomes attractive, especially as the streaming frontier of Sec. 5.2 exposes operating points below one second of CU-LongYAAL that the specialist cascade cannot reach without additional policy machinery on top of the forced aligner.

### Numerical parity as an enabling condition.

The parity bound of Eq. (27), together with the reconstruction-error bound of Eq. (20), is what makes the AlignAtt substrate operational under compiled inference. Without bit-identical acceptance decisions, every ablation against a reference Transformers forward would conflate policy effects with capture noise; with them, every experiment in Sec. 5 is carried out on the engine-native path used in deployment.

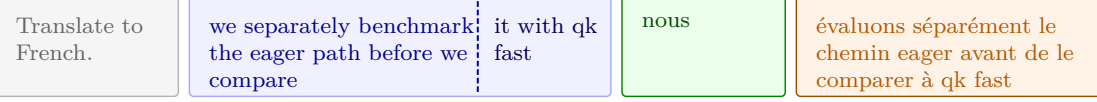
**Limitations and future work.** Our evaluation is single-GPU and single-language-family (en source, one each of de/zh/it target); the policy has not been stress-tested on code-switching or long overlapping speakers. The cascade loop is synchronous: ASR and MT chunks are serialized, so the CU-LongYAAL numbers in Table 4 embed a full chunk round-trip rather than a pipelined one. Finally, the en-zh quality gap suggests that the next lever is on the MT side (larger base model, prompt engineering with target-language priors), not on AlignAtt. AlignAtt itself, as a policy substrate, is not the bottleneck on any pair tested.

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### Word-level selective reconstruction on a live MT draft

target=French • gate=alignatt:source\_frontier • blocked at source word 13 (*fast*)  
system source accepted target prefix



Draft-to-source attention (word level)

Per-word provenance

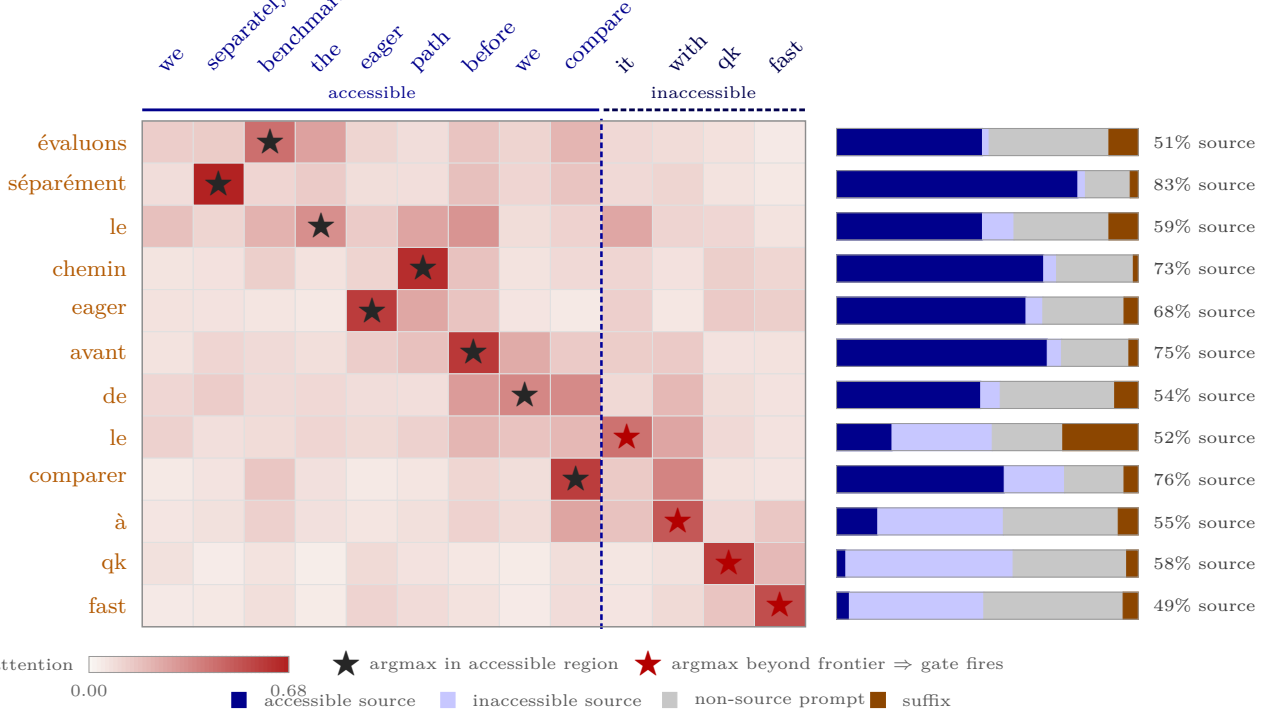
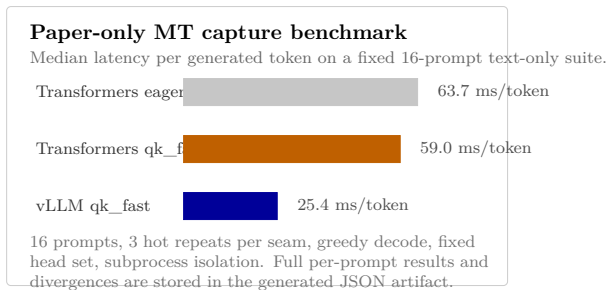


Figure 4: **Word-level selective reconstruction on a live MT draft.** *Top ribbon:* the decoder-only prompt is split into the system instruction, the live source (with the accessibility frontier shown in-line between the committed accessible prefix and the still-inaccessible tail), the already accepted target prefix reused from earlier streaming steps, and the draft currently under inference. *Left panel:* reconstructed draft-to-source attention from the selected AlignAtt heads, aggregated to the word level; rows are drafted words, columns are source words split by the dashed frontier. A star marks the per-row argmax (black when it lands in the accessible region, red when it lands beyond the frontier, which is exactly what triggers the SOURCE-FRONTIER gate). *Right panel:* for each drafted word the reconstructed attention mass is partitioned into the four provenance classes; the numeric annotation is the total source share (accessible + inaccessible). The green band is previously committed target text reused as causal context; it is not itself the object of the gate. The first black stars in the orange draft show that newly proposed words may still align to multiple source words inside the current accessible span, and only the first red star beyond the frontier triggers rejection.



**Figure 5: Inference-time comparison of MT capture seams.** We compare a minimal Transformers eager reference, a Transformers qk-fast reference that reconstructs source rows from captured layer inputs, and the engine-native vLLM qk-fast path used by the shipped backend. The main plot reports the median latency per generated token on a fixed 16-prompt benchmark suite, with one unmeasured warmup pass and repeated hot runs per seam.